Aircraft Closed-Loop Dynamic System Identification in the Entire Flight Envelope Range Based on Deep Learning

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Abstract: To solve the aircraft dynamics modeling problem in the entire envelope range, this work proposes a closed-loop system identification method based on deep learning. A closed-loop flight test was designed, under the framework of the closed-loop flight test, the motion mode of the aircraft was fully stimulated by the input signal of the control rudder surface and the airspeed and position commands. The lateral and longitudinal aerodynamic coefficients were solved from the flight test data, and the black box relationship between the aerodynamic coefficients and their influencing factors was established based on the deep network technology. The aerodynamic coefficient black box model was combined with the dynamics and kinematic equations of the aircraft to form a deep network dynamic model of the aircraft, which belongs to a gray box dynamics model. The deep network can easily and uniformly process different batches of flight test data, thus combining the flight test data at different flight state points, and finally building a complete aerodynamic model within the entire envelope range. Three groups of flight tests were performed: the first group of tests was used for model set training, the second group of test data was used for the selection of the best model, and the third group of flight tests was used for model validation. The model verification was completed from two aspects: the prediction of the aerodynamic coefficient and the prediction of the flight state variables. The results show that the deep network model can complete high-precision modeling of aerodynamic coefficients; and the gray box dynamic model can complete the modeling of aircraft dynamics within the entire envelope, and can be used as a long-term, high-precision flight simulator. **DOI: 10.1061/JAEEEZ.ASENG-5657.** © *2024 American Society of Civil Engineers.*

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Introduction

Aerodynamic parameters and dynamics model of aircraft are of crucial importance in flight simulation, flight control, test flight monitoring, and fault diagnosis. There are generally three methods to obtain aircraft aerodynamic model: theoretical derivation, wind tunnel test, and system identification method based on flight test. Among them, the system identification method directly obtains the aircraft model from flight test data and can obtain the most intuitive and accurate results (Morelli and Klein 2016).

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Dynamic system identification methods can be divided into frequency domain and time domain identification methods. System identification in the frequency domain is a powerful method to obtain the linear dynamics model of the system by estimating the frequency response of input and output based on spectrum analysis and using nonlinear optimization techniques to fit the frequency response by transfer function or state space (Tischler and Remple 2012; Klyde and Schulze 2020; Grauer and Boucher 2020). Linear and nonlinear models can be obtained by adopting different model structures and algorithms for time domain identification, for example, the state space model (Jategaonkar 2015) and nonlinear motion equation model (Simmons et al. 2019; Qing et al. 2023; Hui and Chen 2023) can be obtained by adopting the equation error method, output error method, Kalman filter method, and so on, and the block-oriented model can be obtained by adopting the learning-based method, including the Hammerstein model, Wiener model, Hammerstein-Wiener model (Roudbari and Saghafi 2017; Zhao and Chen 2012), and recurrent neural network model (Roudbari and Saghafi 2014a; Jackson et al. 2020). Among them, the block-oriented nonlinear model and recurrent neural network model can directly establish the black box model between input and output variables, which can be used for state variables' prediction. There are certain limitations in the models built by the preceding methods. The frequency domain method can only get the linear model; the maximum likelihood identification method and extended Kalman filter method require an appropriate hypothesis model and a large amount of calculation, and are sensitive to the initial value of the parameters to be estimated; for the two black box modeling methods, the block-oriented nonlinear model uses the combination of nonlinear static blocks and linear dynamic blocks to model the dynamic characteristics between inputs and outputs

(Roudbari and Saghafi 2017), and the recurrent neural network relied on its own feedback to model the dynamic characteristics (Roudbari and Saghafi 2014b). These two methods do not take advantage of any prior information of the system, which leads to a complex modeling process and difficulty in generalization, and it is difficult to accurately predict the state of a complex system for a long time.

Modeling the dynamic characteristics of an aircraft becomes difficult and challenging over the entire flight envelope, where changes in altitude and speed cause changes in the control and stability derivatives. However, for flight simulation and flight controller design tasks, it is necessary to build a dynamic model in the entire flight envelope. Model stitching techniques are able to identify different linear models at multiple operating points of the system and stitch them together at other state points using interpolation methods. However, the resulting models are inherently linear and have limited accuracy (Tischler and Tobias 2016; Tobias et al. 2018; Callaghan and Kunz 2021). The multimodel structure is a common structure for large-scale modeling, but when the local model structure is complex and the precision is limited, the precision of the multimodel structure will be limited (Adeniran and El Ferik 2016; Emami and Banazadeh 2019; Emami and Roudbari 2018).

To model the flight dynamics in the entire flight envelope, the flight test within the entire flight envelope must be carried out first. The efficient method is to cover the entire flight envelope by driving the aircraft through speed and altitude commands in a closed-loop flight test. However, closed-loop identification is a difficult problem in the field of aircraft system identification. For flight safety, it is often necessary to keep the control system closed during flight tests, especially for unstable aircraft. This leads to the correlation between the input and output variables, such as the elevator and the angle of attack; the pitch rate will become highly correlated, which will make the time domain equation error method invalid (Jategaonkar 2015). In addition, the controller will directly affect the dynamic relationship between the input and output variables, which will lead to the failure of the black box identification method that directly models the relationship between the input and output variables. Many existing black box identification works, such as recurrent neural network modeling and block-oriented identification methods (Roudbari and Saghafi 2017; Zhao and Chen 2012; Roudbari and Saghafi 2014b), are all carried out in the open-loop case. At present, the most effective method for closed-loop identification is to use the joint input–output method (Berger et al. 2018) method to process the input and output data in the frequency domain to extract the accurate frequency response, but these methods can only establish a linear model near the operating point.

The powerful nonlinear approximation capability of feedforward neural networks has led to a wide range of applications in aircraft dynamics modeling, control, and fault diagnosis (Csáji 2001; Taimoor et al. 2021; Boëly and Botez 2010). It was proved in Csáji (2001) that a multilayer feedforward neural network consisting of a finite number of neurons and an arbitrary activation function is a universal approximator, and a general neural network, especially a multilayer perceptron neural network, can be used as a universal approximator for unknown system or unknown mathematical functions. Verma and Peyada (2020) and Hui and Chen (2023) used different neural networks to improve the output error method and estimate the aerodynamic derivative. Tao and Sun (2019) performed robust optimizations for airfoil and wing via deep learning and a multi-fidelity surrogate-based optimization framework, the multi-fidelity surrogate predictions tended to approach the CFD results as the iteration number increases. He et al. (2020), Millidere et al. (2021), and Li et al. (2022) established a high-precision deep network model for aerodynamic data based on deep forward network and wind tunnel databases with different precisions. Li et al. (2021, 2019) established an unsteady aerodynamic reduced-order model based on deep neural network. The previous work fully illustrates the potential of feedforward neural networks in the field of aircraft aerodynamic modeling.

From the perspective of mechanism modeling, the dynamic characteristics of an aircraft can be described by equations of motion and aerodynamic models (Jategaonkar 2015). The dynamic relationships between aircraft states of motion are in accordance with Newton's and Euler's laws, and there is much established work deriving nonlinear equations of motion to describe changes in the states of motion based on Newton's and Euler's laws (McRuer et al. 2014; Zipfel 2000). If the equations of motion are taken as the prior information, then the problem of aircraft dynamics modeling will be transformed into the modeling of aerodynamic characteristics. Inspired by the previous work, this paper proposes a gray box modeling method for aircraft dynamics that combines deep networks and equations of motion to perform closed-loop identification in the range of the entire flight envelopes based on deep network tools. To fully stimulate the motion modes within the flight envelope of interest, open-loop and closed-loop flight tests were performed separately. The deep networks were used as nonlinear approximators for the aerodynamic coefficients modeling. Based on the deep learning technique, different batches of open-loop and closed-loop flight test data were processed simultaneously, thus avoiding the data collinearity problem in the closed-loop linear model identification and establishing a deep network aerodynamic coefficient database in the entire envelope range.

The proposed identification method can simultaneously process flight test data under different flight conditions instead of identifying multiple local models, solving the problem of aerodynamic coefficients modeling, closed-loop dynamic model identification, and long-term state prediction in the entire envelope range. The rest of this paper is organized as follows: The second section describes the object of identification and the design of the flight test. The third section describes the deep learning algorithm and the gray box dynamics modeling method. The fourth section introduces the identification process of aerodynamic coefficient deep network. The verification of identification model is given in the final section.

Aircraft Description and Flight Test Design

In this work, a fixed-wing aircraft simulation model was taken as an object to conduct flight test and model identification research. The inertia and dimension parameters of the aircraft are as follows: $m = 3.93$ kg, $I_x = 0.0517$ kg $\cdot$ m^2, $I_y = 0.3036$ kg $\cdot$ m^2, $I_z = 0.3432$ kg $\cdot$ m^2, $I_{xz} = 0.0053$ kg $\cdot$ m^2, $S = 0.2645 m^2$, $c = 0.3361$ m, and $b = 0.8908$ m. The aircraft has conventional aerodynamic layout, and the control surfaces include elevator, aileron, and rudder, The engine is mounted along the X-axis of the body.

Simulation Model Description

The state variables and input variables of the simulation model are respectively selected as

$$\mathbf{x} = [V \quad \alpha \quad \beta \quad p \quad q \quad r \quad \phi \quad \theta \quad \psi \quad h \quad x \quad y]^T \tag{1}$$

$$U = [\delta_e \quad T \quad \delta_a \quad \delta_r]^T \tag{2}$$

where V, α, β = airspeed, angle of attack, and sideslip angle; p, q, r = three angular velocity components; ϕ, θ, ψ = three Euler attitude angles, roll, pitch, and yaw; h, x, y = altitude and position in the x- and y-directions of Earth axes; and T, δ_e, δ_a, δ_r = thrust and

J. Aerosp. Eng., 2024, 37(6): 04024079

deflection of elevator, aileron, and rudder, respectively. The differential of the state variables of the aircraft is described by the equation of motion. Applying Euler's and Newton's laws, the dynamic equations of translation and rotation can be obtained as Eqs. (3) and (4)

$$\dot{u} = rv - qw - g\sin\theta + \frac{F_x}{m} + \frac{T}{m}$$
$$\dot{v} = pw - ru + g\cos\theta\sin\phi + \frac{F_y}{m}$$
$$\dot{w} = qu - pv + g\cos\theta\cos\phi + \frac{F_z}{m} \quad (3)$$

$$\dot{p} = (c_1 r + c_2 p)q + c_3 L + c_4 N$$
$$\dot{q} = c_5 pr - c_6(p^2 - r^2) + c_7 M$$
$$\dot{r} = (c_8 p - c_2 r)q + c_4 L + c_9 N \quad (4)$$

where u, v, w = component of the airspeed on the body axis; V, α, β can be easily solved from u, v, w; $c_1 = ((I_y - I_z)I_z - I_{xz}^2)/(I_x I_z - I_{xz}^2)$; $c_2 = (I_x - I_y + I_z)I_{xz}/(I_x I_z - I_{xz}^2)$; $c_3 = I_z/(I_x I_z - I_{xz}^2)$; $c_4 = I_{xz}/(I_x I_z - I_{xz}^2)$; $c_5 = (I_z - I_x)/I_y$; $c_6 = I_{xz}/I_y$; $c_7 = 1/I_y$; $c_8 = (I_x(I_x - I_y) + I_{xz}^2)/(I_x I_z - I_{xz}^2)$; and $c_9 = I_x/(I_x I_z - I_{xz}^2)$.

The relationship between Euler angles and angular velocity components can be expressed as

$$\dot{\phi} = p + (r\cos\phi + q\sin\phi)\tan\theta$$
$$\dot{\theta} = q\cos\phi - r\sin\phi$$
$$\dot{\psi} = \frac{1}{\cos\theta}(r\cos\phi + q\sin\phi) \quad (5)$$

The navigation equations describing the displacement velocity of the aircraft centroid are

$$\dot{x} = u\cos\theta\cos\psi + v(\sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi) + w(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)$$
$$\dot{y} = u\cos\theta\sin\psi + v(\sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi) + w(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)$$
$$\dot{h} = u\sin\theta - v\sin\phi\cos\theta - \cos\phi\cos\theta \quad (6)$$

where F_x, F_y, F_z = component of aerodynamic force acting on the aircraft centroid; and L, M, N = roll, pitch, and yaw moments. Aerodynamic force and torque are calculated from dynamic pressure and dimensionless aerodynamic coefficients

$$F_x = QS_w C_x, F_y = QS_w C_y, F_z = QS_w C_z, L = QS_w bC_l, M = QS_w cC_m, N = QS_w bC_n \quad (7)$$

where C_X, C_Y, C_Z, C_m, C_l, C_n = dimensionless coefficients corresponding to the X-, Y-, and Z-axis component of aerodynamic force and the pitching moment, rolling moment, and yawing moment, respectively; these parameters were derived from the discrete data table at a specific state point.

Simulation Flight Test Design

The purpose of this work was to model the dynamic characteristics of aircraft within the entire flight envelope. The selected flight envelope is shown in Fig. 2 with the airspeed range from 30 to 110 m/s and the flight altitude range from 500 to 2,000 m. The closed-loop flight tests were performed, and the altitude and speed control laws were designed by the simulation model. The control laws are shown in Eqs. (8) and (9)

$$\Delta\delta_e = K_z^\theta\theta + K_z^{\dot{\theta}}\dot{\theta} + K_z^h(h - h_g) + K_z^{\dot{h}}\dot{h} \qquad \Delta\delta_T = K_z^V(V - V_g) \quad (8)$$

$$\delta_a = I_{\dot{\phi}}\dot{\phi} + I_\phi\phi + I_\psi(\psi - \psi_g) + I_y(y - y_g) \qquad \delta_r = K_{\dot{\psi}}\dot{\psi} - K_\phi\phi \quad (9)$$

In the framework of closed-loop identification, the motion of the aircraft was excited in two ways. The first way was to drive the aircraft to perform large-amplitude maneuvers by altitude command, speed command, and position command. The second was the application of computer-generated input commands at the control surfaces. Two types of flight tests were designed in this work.

Open-Loop Small-Perturbation Flight Test

To highlight the influence of flight state variable (such as angle of attack, sideslip angle, and angular velocity components) on the aircraft dynamics, small-perturbation flight tests were carried out at different flight test points. The multistep input signal applied directly to the aircraft control surface without the feedback control system operating is an effective input signal to excite small-perturbation motion (Simmons et al. 2019). Alternate multistep elevator, aileron, and rudder multistep inputs are designed as shown in Fig. 1(a).

Closed-Loop Large-Amplitude Maneuvers

To give prominence to the influence of speed and altitude on the dynamic characteristics of the aircraft, the altitude command, speed command, and position command were used to excite large-amplitude maneuvers, such as climbing and descending, acceleration and deceleration, and lateral deviation. The trajectory of the large-amplitude maneuvers should cover the flight envelope of interest as much as possible, as shown in Fig. 2.

Orthogonal Optimized Multisine Inputs to Supplement Information of Flight Test Data

The multisine signals composed of different harmonic components contain specific frequency band ranges. The multisine signals can quickly stimulate the small-amplitude maneuver at different modal frequencies for the aircraft in flight without changing the nominal flight trajectory, the orthogonal property of which is important to achieve the low correlation required for the explanatory variables (Jategaonkar 2015; Morelli 2012). To this end, in the process of both the open-loop small-perturbation flight test and closed-loop large-amplitude maneuvers, orthogonal multisine signals were applied to three flight control surfaces simultaneously to excite the aircraft motion more adequately without deviating from the nominal trajectory for large-amplitude maneuvers. Each multisine signal u_i is superimposed by n_i sinusoidal signals with different frequencies and phases

$$u_i = \sum_{k\in\{1,2,\ldots,m\}} A_{jk}\sin\left(\frac{2\pi kt}{T} + \phi_{jk}\right), \quad j = 1,2,\ldots,n_i, i = 1,2,\ldots,N \quad (10)$$

where T = excitation period; $\omega_k = 2\pi k/T, k \in \{1,2,\ldots,m\}$ represents the frequency component of the harmonic signal contained in the multisine signal u_i; and the amplitude and phase of each harmonic signal are A_{jk} and ϕ_{jk}. The value of A_{jk} is determined according to Eq. (11), where A is the expected amplitude of the

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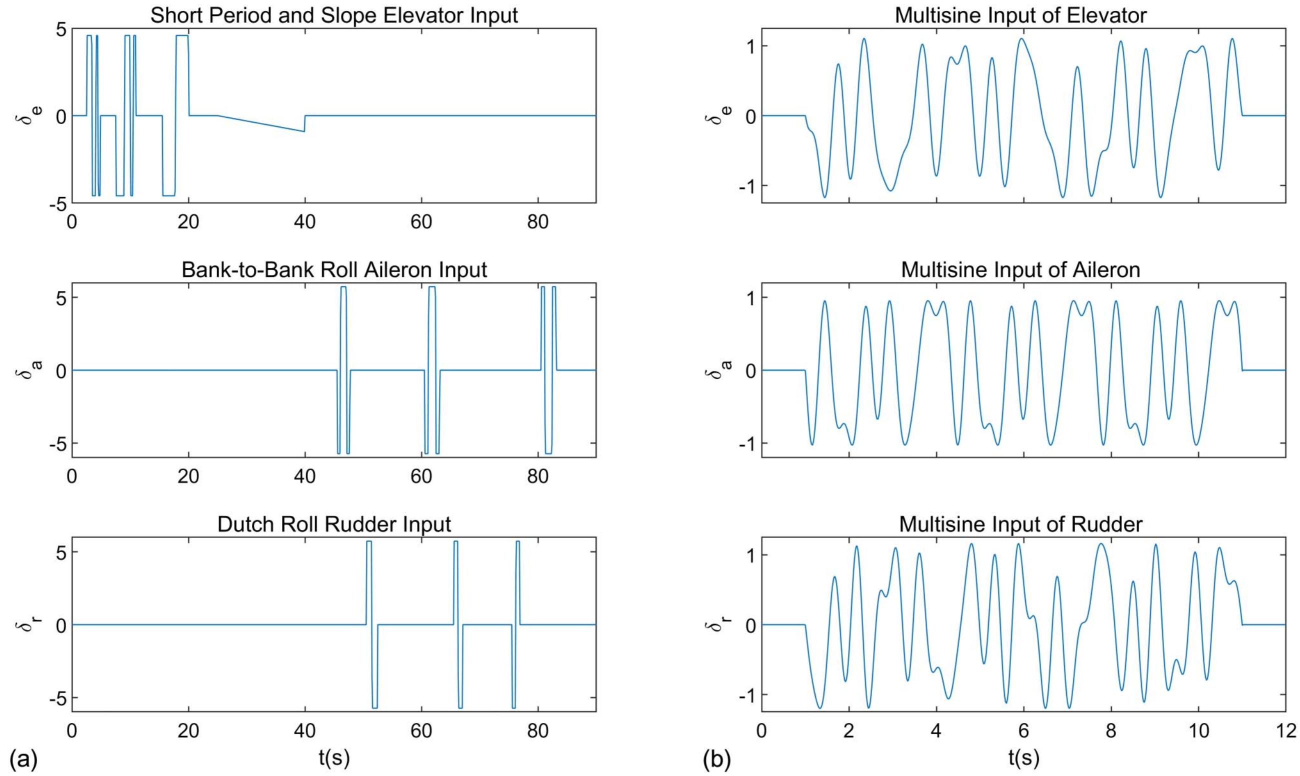

Fig. 1. Multistep input and multisine input signals of flight control surface applied throughout open- and closed-loop flight test maneuvers.

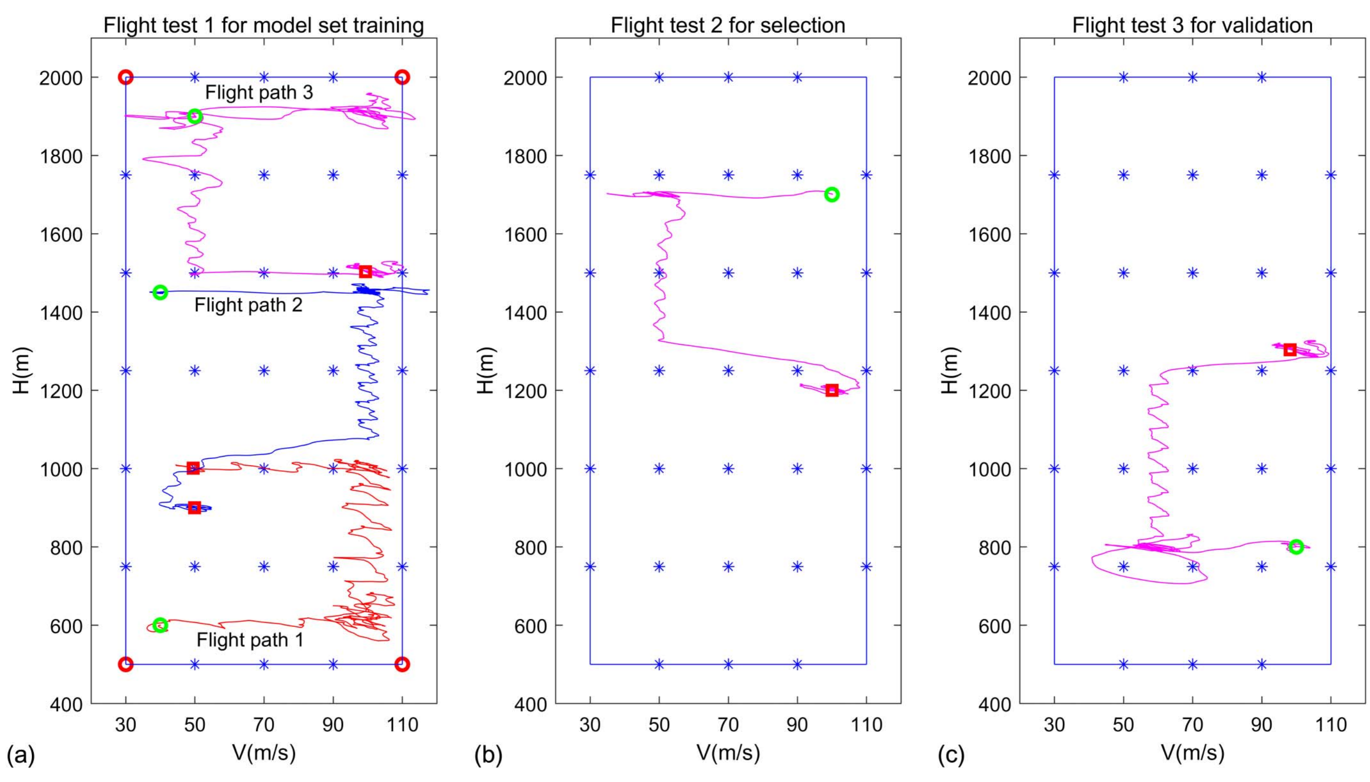

Fig. 2. Selected flight envelope range and flight path of the three groups of flight test maneuvers.

signal u_i. To avoid the situation that the amplitude of u_i is too large, the phase ϕ_{jk} of each sinusoidal signal is optimized by minimizing the relative peak factor (RPF). In this research, the excitation period of the designed multisine signal for elevator, aileron, and rudder was $T = 10$ s, the designed frequency band was [0.2, 2.1] Hz, and each multisine signal was composed of seven harmonic components; the frequencies (in Hz) of the harmonic components contained in the three signals are given in Table 1

Table 1. The frequencies (in Hz) of the harmonic components contained in different multisine signals

Multisine signal	Frequency component
Frequency component $\omega k1$ for elevator multisine signal	$2/T$, $5/T$, $8/T$, $11/T$, $14/T$, $17/T$, $20/T$
Frequency component $\omega k2$ for aileron multisine signal	$3/T$, $6/T$, $9/T$, $12/T$, $15/T$, $18/T$, $21/T$
Frequency component $\omega k3$ for rudder multisine signal	$4/T$, $7/T$, $10/T$, $13/T$, $16/T$, $19/T$, $22/T$

$$A_{jk} = \frac{A_i}{\sqrt{n_i}}, \quad i = 1, 2, \ldots, N \tag{11}$$

$$\mathrm{RPF}(u_i) = \frac{\max(u_i) - \min(u_i)}{2\sqrt{2}rms(u_i)}, \quad i = 1, 2, \ldots, N \tag{12}$$

The phase ϕ_{jk} of each harmonic was optimized by using the simplex method to minimize the RPF of the multisine signal u_i. The periodicity and orthogonality of multisine signals enable the three signals to be applied simultaneously and continuously. The multisine signals of the elevator, aileron, and rudder were designed as Fig. 1(b). During the open-loop and closed-loop flight tests described in the sections "Open-Loop Small-Perturbation Flight Test" and "Closed-Loop Large-Amplitude Maneuvers," the multisine signals shown in Fig. 1(b) are simultaneously superimposed on the three control surfaces to more fully excite the motion modes of different frequencies of the aircraft, and the amplitude of the multisine input signal is set to the maximum value that does not affect the nominal flight trajectory of the flight test.

Flight Tests for Aerodynamic Model Identification, Selection, and Validation

A total of three groups of flight tests were carried out, which were used for model set identification, optimal model selection, and model verification, respectively. The final flight trajectory is shown in Fig. 2. The first group of flight tests consisted of four small-perturbation flight tests and three large-amplitude maneuvers, and the flight test data generated were used for the identification of the aerodynamic coefficient model set. The flight trajectories of the first group of flight tests are shown in Fig. 2(a), in which open-loop small-perturbation flight tests were performed at the four vertices of the flight envelope marked with circles, and the closed-loop large-amplitude maneuvers were performed along the flight path from the circle to the square. The second and third groups of flight test maneuvers were used for model selection and model verification, respectively, and each included one closed-loop large-amplitude maneuver, as shown in Figs. 2(b and c). As described in the section "Orthogonal Optimized Multisine Inputs to Supplement Information of Flight Test Data," multiple sinusoidal inputs were applied at the control surfaces during the open-loop small-disturbance flight test and the closed-loop test.

For the first group of flight tests consisted of four small-perturbation flight tests and three large-amplitude maneuvers; the cross-plots of aircraft states are shown in Fig. 3. The cross-plots do not show characteristics close to a diagonal lines or ellipse, implying the explanatory variable data from this group of flight tests have very low pairwise correlations.

Nonlinear Flight Dynamics Modeling Based on Deep Networks

In the current work, the aerodynamic coefficient deep network model was constructed based on flight test data, and the aerodynamic coefficient deep network model was combined with the aircraft motion equation to form the gray box dynamics model.

Deep Network Structure and Training Methods

The simplest perceptron contains three operation rules of addition, multiplication, and activation function. The multilayer perceptron network containing multiple perceptrons was classified as feedforward deep neural network (Nielsen 2015), and was composed of

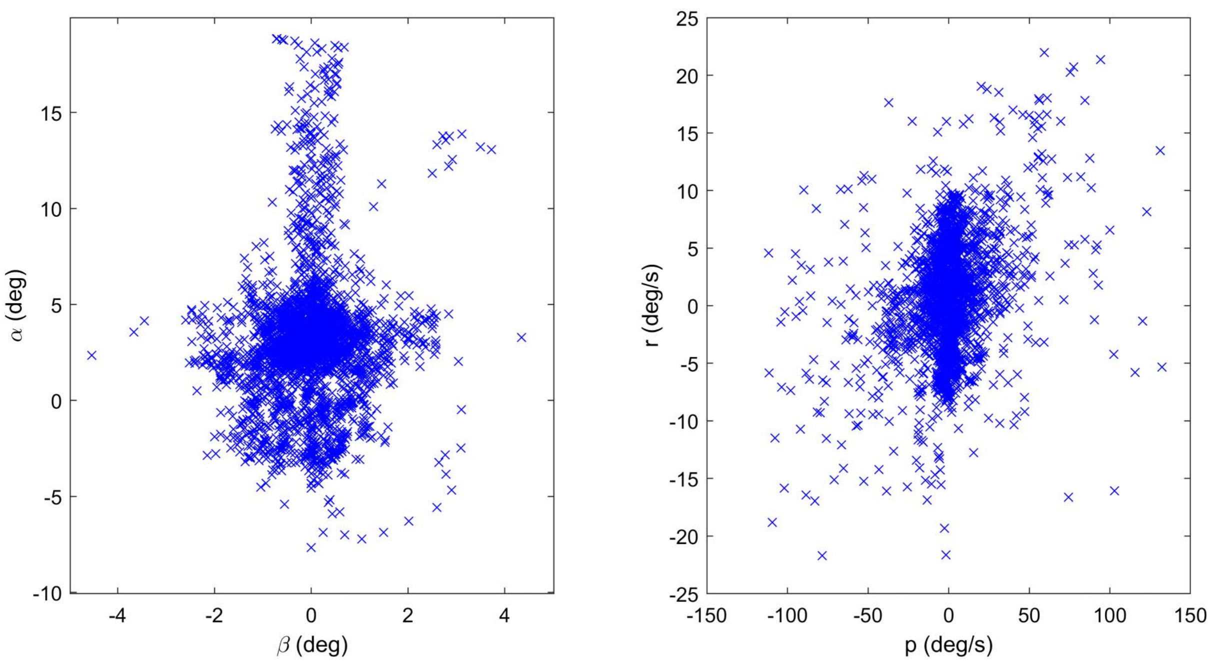

Fig. 3. Explanatory variable coverage for the first group of flight tests.

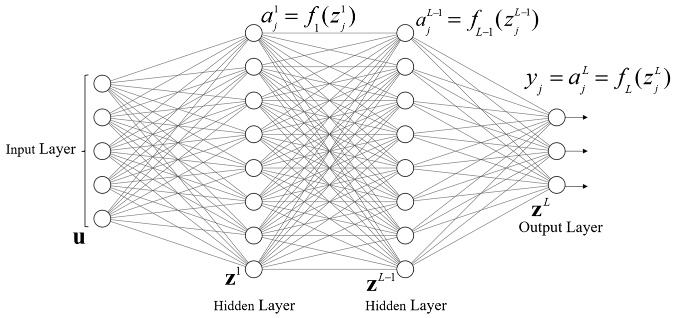

Fig. 4. Structure of multilayer perceptron.

input layer, hidden layer, and output layer, as shown in Fig. 4. The forward propagation process of a neural network containing $L-1$ hidden layers is as follows.

The propagation from the input layer to the first hidden layer is shown in Eq. (13), and the output vector of the first hidden layer is $\mathbf{a^1}$. In Eq. (13), a_j^1, the jth component of $\mathbf{a^1}$, is the output of the jth neuron of the first hidden layer; $\mathbf{W^1}$ is the weight matrix between the input layer and the first hidden layer; b^1 is the bias vector of the first hidden layer; and f_1 is the activation function of the first hidden layer

$$\begin{aligned} \mathbf{z^1} &= \mathbf{W_1 u} + \mathbf{b^1} \\ a_j^1 &= f_1(\mathbf{z}_j^1) \end{aligned} \tag{13}$$

Similarly, the output of each neuron in the lth hidden layer is a_j^l, and the output of each neuron in the output layer is y_j. The formulas for a_j^l and y_j are as Eqs. (14) and (15), where $\mathbf{a^{l-1}}$ and $\mathbf{a^{L-1}}$ are the output vector of the $(l-1)$th and $(L-1)$th hidden layer, respectively

$$\begin{aligned} \mathbf{z^l} &= \mathbf{W^l a^{l-1}} + \mathbf{b^l} \\ a_j^l &= f_l(z_j^l) \end{aligned} \tag{14}$$

$$\begin{aligned} \mathbf{z^L} &= \mathbf{W}^L \mathbf{a}^{L-1} + \mathbf{b}L^\mathbf{1} \\ y_j &= a_j^L = f_L(z_j^L) \end{aligned} \tag{15}$$

Activation functions were selected as hyperbolic tangent function in the hidden layers and purelin function in the output layer, as in

$$\begin{aligned} f_l(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \qquad l = 1, 2, \ldots, L-1 \\ f_L(x) &= x \end{aligned} \tag{16}$$

At the tth discrete input-output data pair, the local cost function is chosen as the sum of squares of the network output and the measured response vector

$$E(t) = \frac{1}{2}[\mathbf{y}_m(t) - \mathbf{y}(t)]^T[\mathbf{y}_m(t) - \mathbf{y}(t)] = \frac{1}{2}\mathbf{e}^T(t)\mathbf{e}(t) \tag{17}$$

where $y_m(t)$ = measured output vector of the actual system; $\mathbf{y}(t)$ = output vector of the deep network; and E = error between the network output and the actual measured value.

For deep network training, it is necessary to calculate the gradient of the cost function to the weights and bias vectors of each layer

$$\frac{\partial E}{\partial W_{jk}^l} = \frac{\partial E}{\partial z_j^l} a_k^{l-1} \tag{18}$$

$$\frac{\partial E}{\partial b_j^l} = \frac{\partial E}{\partial z_j^l} \tag{19}$$

where $(\partial E)/(\partial \mathbf{z^L})$ is calculated first as formula (20), where $\odot$ corresponds to the elementwise product of the two vectors

$$\frac{\partial E}{\partial \mathbf{z^L}} = \left(\frac{\partial E}{\partial \mathbf{a^L}}\right) \odot f_L'(\mathbf{z^L}) \tag{20}$$

$(\partial E)/(\partial \mathbf{z^l})$ is then calculated by $(\partial E)/(\partial \mathbf{z^{l+1}})$ and the backpropagation rule, as in

$$\frac{\partial E}{\partial \mathbf{z^l}} = (\mathbf{W^{l+1}})^T \left(\frac{\partial E}{\partial \mathbf{z^{l+1}}}\right) \odot f_l'(\mathbf{z^l}) \tag{21}$$

Combining to Eqs. (18)–(21), the gradient of the cost function to the weights and bias vectors of all hidden layers can be derived.

The Adam algorithm, an efficient stochastic optimization method that only requires the first-order gradient, is applied for deep neural network training. The method is easy to implement, computationally efficient with little memory requirement, and well suited to problems with large data and/or parameters (Kingma and Ba 2014). In the task of modeling aerodynamic coefficients, in order to contain as much dynamic information as possible, the amount of flight test data is large. In this case, the Adam algorithm is very suitable. The steps of parameter update at the tth time are as follows:

1. Calculate $g_t = \nabla_\theta E(\theta_{t-1})$, the gradient of the cost function $E(t)$ to the parameter θ (weights and bias vector) in the tth optimization process through equation to Eqs. (18)–(21).
2. Update the biased first moment estimate

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \tag{22}$$

3. Update the biased second raw moment estimate

$$\begin{aligned} v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t \odot g_t \\ v_0 &= 0 \end{aligned} \tag{23}$$

4. Calculate the bias-corrected first moment estimate

$$\begin{aligned} \hat{m}_t &= m_t/(1-\beta_1^t) \\ m_0 &= 0 \end{aligned} \tag{24}$$

5. Compute the bias-corrected second raw moment estimate

$$\hat{v}_t = v_t/(1-\beta_2^t) \tag{25}$$

6. Update parameters

$$\theta = \theta_{t-1} - \alpha \cdot \hat{m}_t \Big/ \left(\sqrt{\hat{v}_t} + \varepsilon\right) \tag{26}$$

where $\alpha, \beta_1, \beta_2, \varepsilon$ = adjustable parameters, and are set to default values in this work: $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$.

Deep Network–Based Aerodynamics Coefficient Model and Gray Box Dynamics Model

It can be known from prior knowledge of flight dynamics that the aerodynamic coefficients are related to the deflection of the control surface, the airspeed, the airflow angle, and the Euler angle rate; the longitudinal aerodynamic coefficients C_X, C_Z, C_m are the algebraic function of the elevator deflection, angle of attack, sideslip angle, airspeed, and pitch rate; lateral aerodynamic coefficients C_y, C_l, C_n are algebraic functions of rudder deflection, aileron deflection, angle of attack, sideslip, airspeed, roll, and yaw rate; and these quantities are measurable in flight tests. The measurements in a flight test include elevator, rudder, aileron deflection, airspeed, angle of attack, sideslip angle, pitch, roll, yaw angle and angular rate, and acceleration a_x, a_y, a_z. Random noise is added to the measurements to achieve a signal-to-noise ratio of about 10, so the measurements need to be locally smoothed in the time domain before being used for modeling (Morelli and Klein 2016). The aerodynamic coefficients $C_X, C_Z, C_m, C_y, C_l, C_n$ cannot be measured directly, but can be calculated by inversely solving formulas (3) and (4), for example, by inversely solving Eq. (4), C_l, C_m, C_n can be obtained as Eq. (27). In the formula, $\dot{p}, \dot{q}, \dot{r}$ are obtained from the numerical differentiation of p, q, r, and the other variables can be directly measured. The values of C_X, C_Z, C_y can be derived in the same way. Next, we adopt the deep network shown in Fig. 4 to model the algebraic function relationship between the longitudinal and lateral aerodynamic coefficients and their influencing factors, and treat the longitudinal and lateral aerodynamic coefficients separately, that is, the modeling variables of the longitudinal and lateral aerodynamic coefficients $\mathbf{u}_{\text{longitudinal}}$ and $\mathbf{u}_{\text{lateral}}$ are shown in Eqs. (28) and (29) respectively

$$\begin{aligned} C_l &= (I_x\dot{p} - I_{xz}\dot{r} + (I_z - I_y)qr - I_{xz}pq)/(QS_wb) \\ C_m &= (\dot{q} - c_5pr + c_6(p^2 - r^2))/(c_7QS_wc) \\ C_n &= (I_z\dot{r} - I_{xz}\dot{p} + (I_y - I_x)pq + I_{xz}qr)/(QS_wb) \end{aligned} \tag{27}$$

$$\begin{aligned} \mathbf{u}_{\text{longitudinal}} &= [\,V \quad \alpha \quad \beta \quad q \quad \delta_e\,]^T \\ \mathbf{y}_{\text{longitudinal}} &= [\,C_x \quad C_z \quad C_m\,]^T \end{aligned} \tag{28}$$

$$\begin{aligned} \mathbf{u}_{\text{lateral}} &= [\,V \quad \alpha \quad \beta \quad p \quad r \quad \delta_a \quad \delta_r\,]^T \\ \mathbf{y}_{\text{lateral}} &= [\,C_y \quad C_l \quad C_n\,]^T \end{aligned} \tag{29}$$

The training data are mixed of the closed-loop test data and the open-loop test data of Flight Test 1. Taking $\mathbf{u}_{\text{longitudinal}}$ as an example, the correlation coefficients between the modeling variables of closed-loop test data and mixed data are given in Table 2, where the values not in bold represent the correlation of closed-loop test data and the values in bold represent the correlation of mixed data. It can be seen that the correlation between modeling variables of mixed data is significantly lower than that of closed-loop test data. For example, for airspeed and angle of attack, the correlation coefficient of closed-loop data is 0.8709 and that of open-loop test data is 0.3226, indicating that the correlation of mixed data is significantly reduced, which is conducive to the aerodynamic coefficients modeling.

For the aircraft aerodynamic coefficient modeling task, the flight test data used for network training contain N sampling points. When N is relatively large, to save training memory and obtain faster training speed, the minibatch training method (Nielsen 2015) is adopted. It is to randomly divide N pairs of input and output samples into n subsets, with each subset containing m pairs of samples. Calculate the average gradient of m pairs of samples on each subset, and update the parameters based on the Adam algorithm as an iteration. It takes n iterations to traverse n subsets, as an epoch. The aerodynamic coefficient deep network training finally goes through T epochs. Specifically, the settings used for aerodynamic

Table 2. Correlation coefficients between the modeling independent variables $\mathbf{u}_{\text{longitudinal}}$

Modeling variable	V	α	q	δ_e	β
V	1	**0.3226**	**0.0819**	**0.1797**	**0.0838**
α	0.8709	1	**0.2130**	**0.2311**	**0.0128**
q	0.0381	0.3068	1	**0.3448**	**0.0137**
δ_e	0.4550	0.5876	0.6026	1	**0.0067**
β	0.0027	0.0121	0.0158	0.0161	1

Note: Values not in bold represent the closed-loop test data. Values in bold represent mixed data.

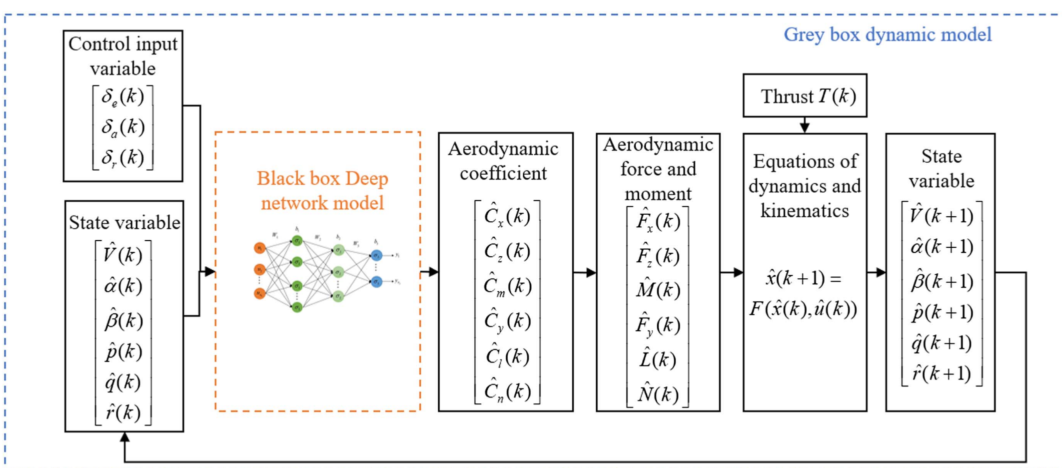

Fig. 5. Relationship between the black box aerodynamic coefficients model and the gray box dynamic model.

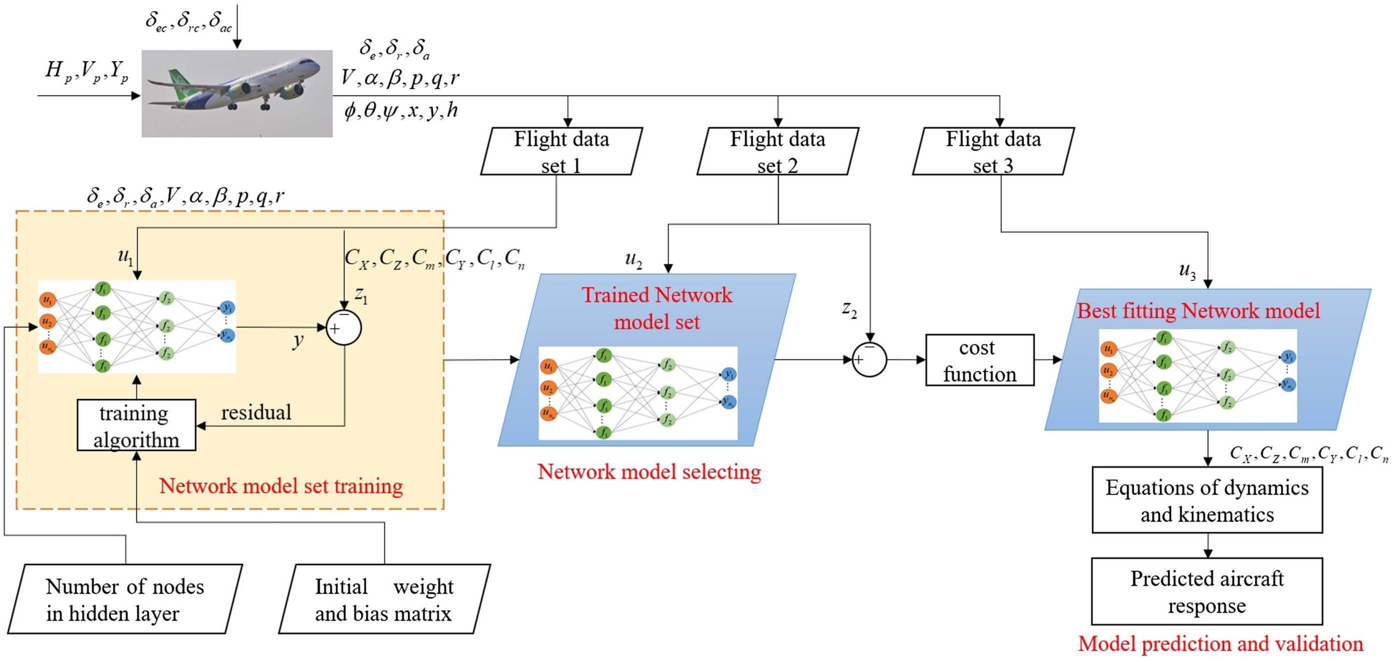

Fig. 6. Model set training, model selection, and validation process. (Image by authors.)

model training in this work were $N = 48{,}000$, $m = 128$, $n = 375$, and $T = 500$.

Deep network is used to model the aerodynamic coefficient of the aircraft. The dynamic relationship between the motion states of the aircraft is described by the equation of motion in Eqs. (3)–(6), and Eq. (30) represents its discrete form. The black box aerodynamic coefficient model is combined with the discrete-form white box equation of motion to constitute a gray box model describing the dynamic characteristics of the aircraft. The relationship between black box aerodynamic coefficient model and gray box dynamic model is shown in Fig. 5

$$\hat{x}(k+1) = F(\hat{x}(k), \hat{u}(k)) \tag{30}$$

Deep Network Aerodynamic Model Set Training and Selection

The performance of a deep network is closely related to the network structure. The size of the neural network and the type of network topology are critical to the training and generalization of the neural network. Larger networks train faster but may have overfitting that leads to their poor generalization, whereas smaller networks train more slowly but may generalize better (Nielsen 2015). At present, there are not effective and direct methods to optimize the structure of the neural network. Although some works propose to apply the genetic algorithm to optimize the hyperparameters of the neural network, when the amount of data is large, it takes a long time for a single network training. Algorithms to find the best network structure take a lot of time. Therefore, this paper still adopted the experimental method to determine the optimal network structure.

As mentioned in the section "Flight Tests for Aerodynamic Model Identification, Selection, and Validation," a total of three groups of flight tests were designed to collect three sets of flight test data, of which the first set was used for model set training, that is, training multiple deep networks with different numbers of nodes and different topologies to form a model set. The second set was applied for the selection of the model. The third set was used for model validation. The model set training and selection process is shown in Fig. 6.

The generalization performance of a deep network is expressed in terms of mean-square error (MSE), as shown in Eq. (31), where $\mathbf{y}_m$ represents the network prediction vector and $\mathbf{y}$ represents the corresponding experimental measurement, N represents the length (number) of training or test samples, y_{ms}, y_s represents the sth sample of y_m and y. The smaller the mean-square error, the better the prediction effect

$$\mathrm{MSE} = \frac{1}{N}\sum_{s=1}^{N}(y_{ms} - y_s)^2 \tag{31}$$

Uzair and Jamil (2020) proposed that usually implementing three numbers of hidden layers gives the optimal performance in terms of time complexity and accuracy, and also unnecessary increment in the neurons or layer will led to overfitting problem. Therefore, in this paper, two and three hidden layers were considered in aerodynamic coefficient modeling. The number of network nodes increased gradually, and a total of six network structures were trained, as listed in Table 3.

Based on the first set of flight test data, different aerodynamic coefficient network models were trained utilizing Adam algorithm after 500 training epochs in each case in Table 3, thereby obtaining a set of models. The predicted performance was evaluated by the MSE between the prediction of network models and the test

Table 3. Six models of deep network used for aerodynamic coefficients modeling

Model	Number of hidden layers	Number of nodes in each hidden layer
Model 1	2	10, 10
Model 2	2	20, 20
Model 3	3	6, 6, 6
Model 4	3	8, 8, 8
Model 5	3	10, 10, 10
Model 6	3	15, 15, 15

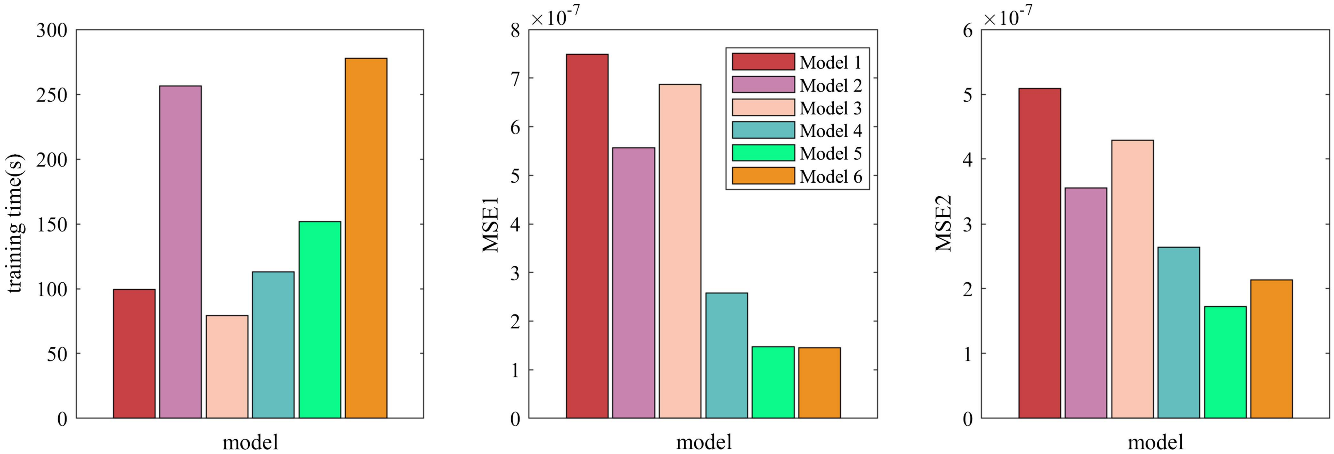

Fig. 7. Training time and MSE values of different deep networks model for the aerodynamic coefficient C_x, MSE1 for Flight Test Data Set 1, and MSE2 for Flight Test Data Set 2.

measurements. The training time of deep networks for aerodynamic coefficient C_x with different structures and the MSE in the training set and test set are shown in Fig. 7. When the training times are all 500, the training time increases significantly with the increase of the depth and width of the deep network, while the training error of the network tends to decrease, and the performance of deep networks with three hidden layers is better than those with two hidden layers. However, when the number of nodes in the three-layer hidden layer network is increased from 10 to 15 (from Model 5 to Model 6), the training error MSE1 does not decrease significantly, while MSE2 increases, indicating that the expansion of network structure causes difficulties in network training and poor generalization. Therefore, Model 5 (three hidden layers with 10 nodes in each layer) was selected as the best model for C_x, that is, the largest network structure with no overfitting phenomenon was selected. Based on the same principle, the optimal model structure of C_z, C_m, C_y, C_l, C_n was selected as three hidden layers, and the number of nodes was determined as 6, 8, 8, 10, and 15, respectively.

Deep Network Model Validation

Model validation works in two ways. On the one hand, we carried out the flight test and model identification work based on the simulation model; because the theoretical value of the aerodynamic coefficient of the simulation model is known, the comparison between the theoretical value of the aerodynamic coefficient and the predicted value of the deep network can directly illustrate the accuracy of the identified deep network of aerodynamic coefficients. On the other hand, the deep network dynamics model can directly predict the flight state variables of the aircraft. Comparing the predicted flight state variables of the deep network dynamics model with the measurements of the simulated flight test can further illustrate the accuracy of the identified deep network model of aerodynamic coefficients.

By changing the flight state variables and inputting the state variables into the simulation aircraft model and the deep network aerodynamic model, respectively, the theoretical values of aerodynamic coefficients and the predicted values of the deep network under different state variables can be obtained.

Fig. 8 shows the variation of the theoretical aerodynamic coefficients of the simulation model and the predicted aerodynamic coefficients of the deep network model with the angle of attack, airspeed, and sideslip angle. For longitudinal aerodynamic coefficient C_x, C_z, C_m, the predicted values of deep network model are very close to the theoretical values. When the angle of attack is small, the prediction accuracy of the deep network is very high, and when the angle of attack is increased to 15°, there will be a small prediction error because the angle of attack data in the flight test data used for network training are mainly concentrated within 15°. It can be seen that the deep network can also learn the influence of the airspeed and sideslip angle on the longitudinal aerodynamic coefficient. It is obvious that compared with the angle of attack, the sideslip angle has less influence on the longitudinal aerodynamic coefficient; however, the deep network can still learn the influence of the sideslip angle on longitudinal aerodynamic coefficient.

Fig. 9 shows the variation of lateral aerodynamic coefficient with sideslip angle, airspeed, and angle of attack. The sideslip angle in Flight Test Data Set 1 is mainly concentrated within 4°. Within the range of ±4°, the lateral aerodynamic coefficient prediction accuracy of the deep network is relatively high. Beyond this range, there will be a small deviation between the prediction result and the theoretical value. As the airspeed and angle of attack change, the lateral aerodynamic coefficient will change slightly. The deep network has learned the variation law of the lateral aerodynamic coefficient with the airspeed and angle of attack. Compared with the sideslip angle, the effect of the angle of attack on the lateral aerodynamic coefficient is less obvious, but the deep network can learn the slight influence of the angle of attack on the lateral aerodynamic coefficient.

The preceding work completes the model set training and model selection for each aerodynamic coefficient, and the respective best model for each aerodynamic coefficient (C_x, C_z, C_m and C_y, C_l, C_n) constitutes the complete aerodynamic coefficients model. Combining the identified deep network model of aerodynamic coefficients with the equation of motion of the aircraft, the deep network dynamics model is obtained. Applying the same control commands as the third group of flight test maneuvers to the deep network dynamic model, namely, altitude command, speed command, and lateral deviation command, the deep network dynamics model predicted the state variables of the aircraft under closed-loop conditions. The comparison between the predicted state variables of the deep network dynamic model and the measurements of the third group of flight test is shown in Figs. 10 and 11.

It can be observed that the predicted values of the aircraft state variables highly match the test measurements during the entire third group of flight test maneuvers, indicating that the deep network

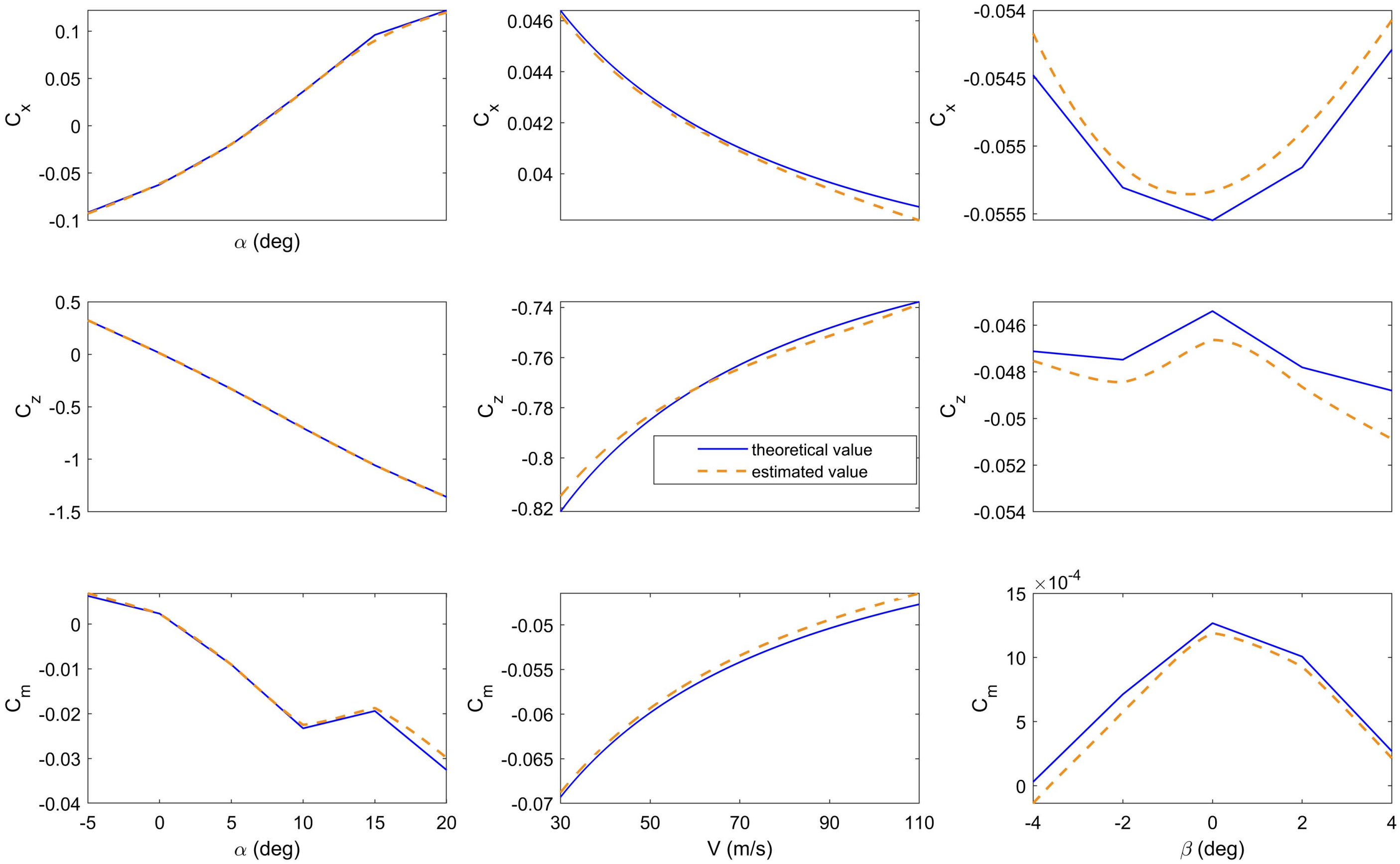

Fig. 8. Theoretical and estimated values of longitudinal aerodynamic coefficients at different angles of attack, airspeeds, and sideslip angles.

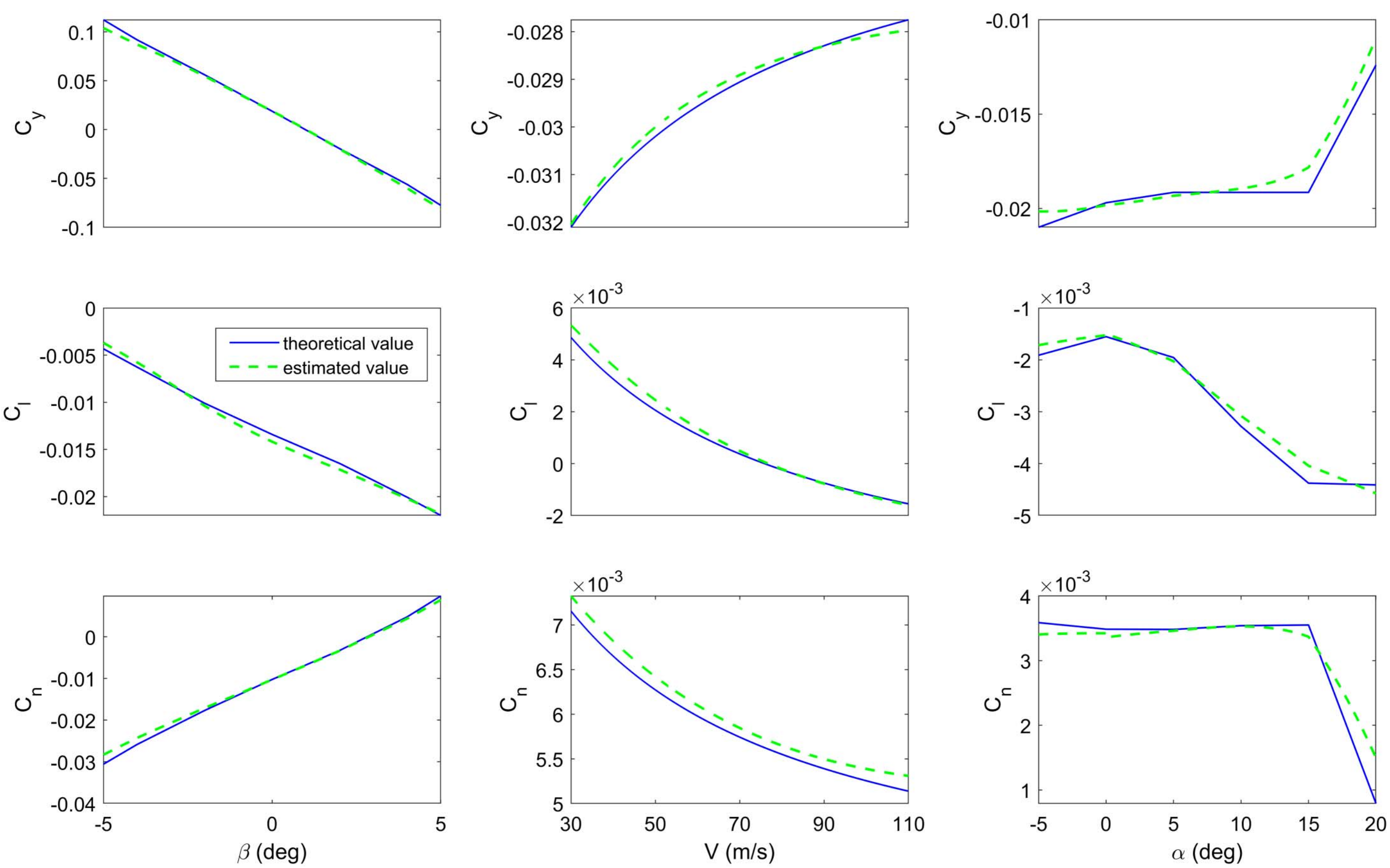

Fig. 9. Theoretical and estimated values of lateral aerodynamic coefficients at different sideslip angles, airspeeds, and angles of attack.

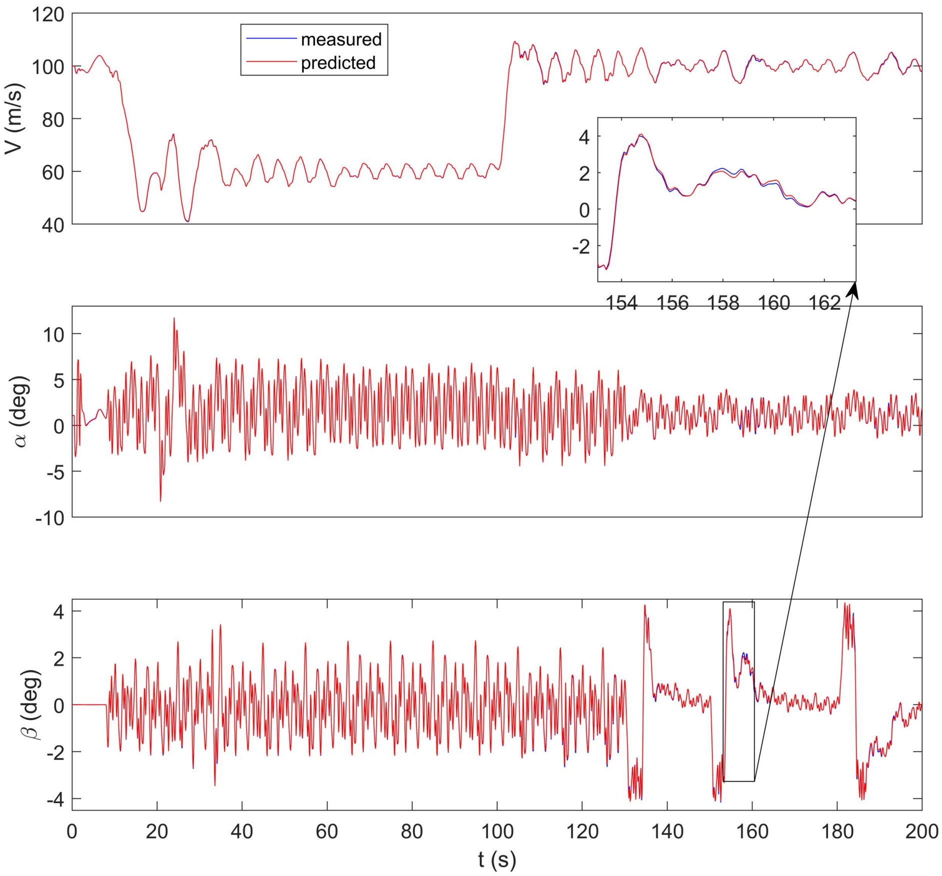

Fig. 10. Measurements of airspeed, angle of attack, sideslip angle, and responses of deep network dynamics model in the third flight test.

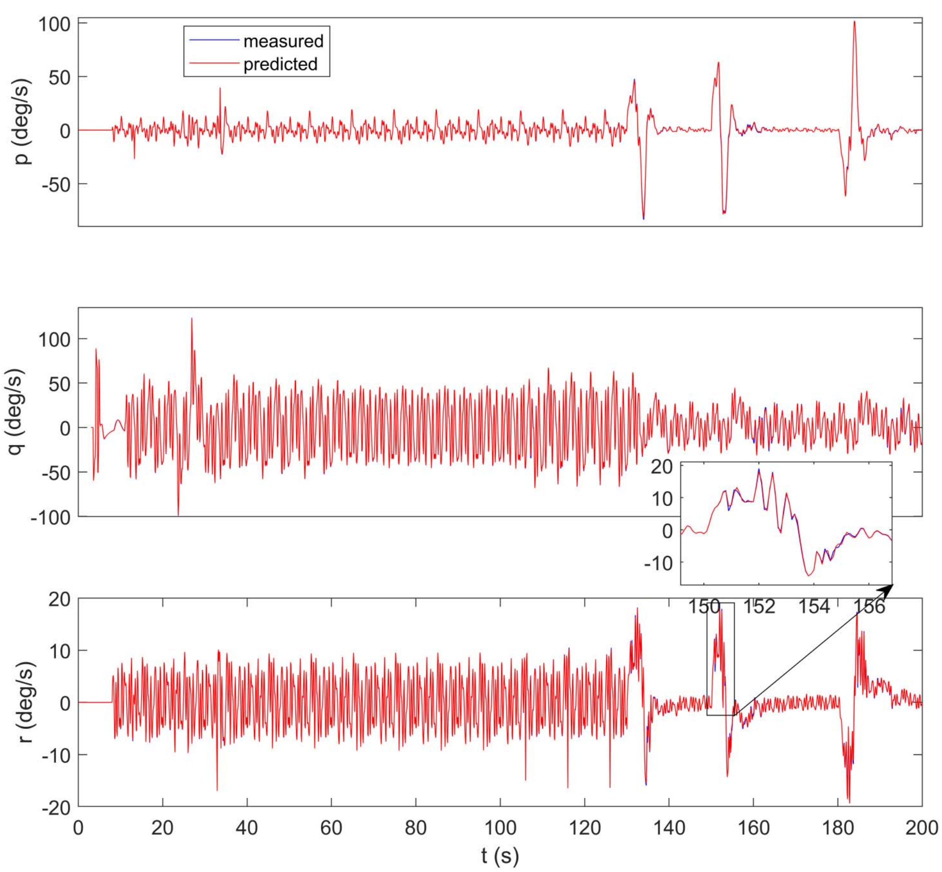

Fig. 11. Measurements of angular velocity components and responses of deep network dynamics model in the third flight test.

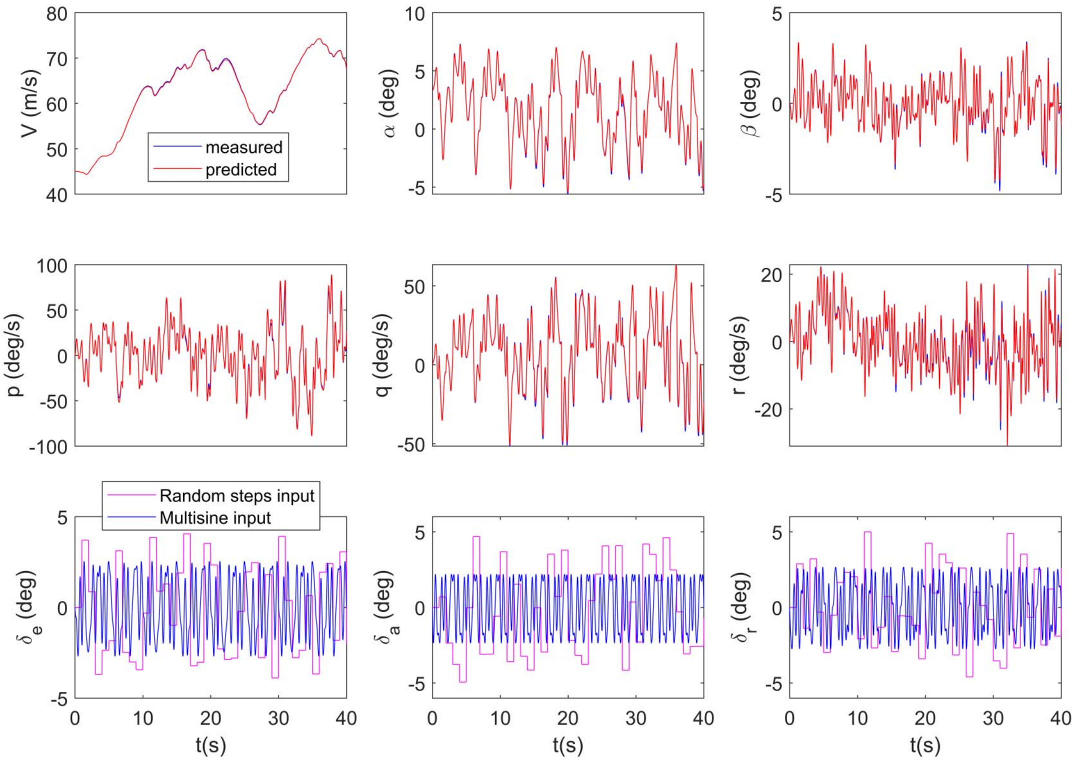

Fig. 12. Input signal, measurements of open-loop test, and the responses of deep network dynamics model.

dynamic model can complete long-term and high-precision prediction of the dynamic response of the aircraft.

Finally, in order to verify the open-loop prediction capability and the generalization performance of the obtained deep network dynamics model, an open-loop flight test was alternately stimulated by random steps input at the airspeed of 45 m/s and the altitude of 1,500 m, and multisine signals were superimposed on the three control surfaces simultaneously. The random input signal is a new signal that did not appear in Flight Tests 1 and 2. The random steps and multisine input signals, along with the measurements of the open-loop flight test and response of the deep network dynamics model, are shown in Fig. 12. It can be seen that the deep network dynamics model can accurately predict the response of the aircraft when excited by random steps and multisine input simultaneously in the open-loop case, which further indicates the generalization ability of the deep network dynamics model.

Conclusion

In this work, the deep network was used to model the aerodynamic coefficients and identify the nonlinear dynamic of the aircraft in the full flight envelope. The main work and conclusions are as follows:

- The open-loop small-perturbation flight test and the closed-loop large-amplitude maneuver flight test with multiple sinusoidal inputs were designed to achieve the low correlation required for the explanatory variables and enrich the information of aerodynamic coefficient modeling. The flight trajectory figures show that the aerodynamic coefficient modeling variables in the flight test data cover a wide range, the explanatory variables converge and the correlation coefficients show low correlation of modeling variables in the mixed flight test data.
- The deep network can model the aerodynamic coefficients of the aircraft without any prior knowledge of the aerodynamic model. The Adam algorithm with high computational efficiency was used for network training, and the training data include both open-loop flight test data and closed-loop flight test data. That is, it is possible to process different batches of flight test data.
- The deep network can accurately predict the aerodynamic coefficients under different conditions, and the deep network can learn the coupling effect between lateral and longitudinal motion.
- The deep network model was combined with the discrete motion equation to form a deep network dynamics model, which can complete the prediction of the aircraft motion state in both closed-loop and open-loop validation flight tests, which illustrates the good generalization performance of the deep network dynamics model.

Data Availability Statement

All data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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